

Local data

Predictors

Research loop

Aggregation

OUTER RESEARCH LOOP

Research language model: fixed weights

Evidence cards + previous trials

1 Propose

Hypothesis +
design identifier

2 Instantiate

Prediction head +
training settings

3 Train + evaluate

Run candidate;
collect diagnostics

4 Retain / revise

Compare development;
record trial outcome

INNER COLLABORATIVE TRAINING

Coordinator: aggregate parameters and development scores

Send shared parameters; receive local updates and evaluation metrics

Lab 1

Scenario A

Train / dev

Local model

Lab 2

Scenario B

Train / dev

Local model

Lab K

Scenario K

Train / dev

Local model

Measurements remain within each laboratory

Candidate design

Development
evidence

Selected

OUTPUTS

Selected design
+ fitted predictor

Research history
all trial results

TASK-SPECIFIC PREDICTORS AND SCORERS

Drug-target interactions

Compound + protein to interaction

Proteomic efficacy

Observed proteome + drug to efficacy

Cell perturbations

Single gene / double gene / drug